

Performance Comparison of YOLO Models Before and After Data Augmentation

This section presents a comparative analysis of the performance of YOLOv5, YOLOv8, YOLOv9, YOLOv10, YOLOv11, and YOLOv12 models for object detection across three classes: car, dog, and person. The evaluation metrics include the F1 score and the precision-recall (PR) curve values for each class. Results are reported for two scenarios: without data augmentation and with data augmentation.

Results Before Data Augmentation

Table 1 summarizes the performance of the YOLO models without data augmentation.

Table 1: Performance Metrics of YOLO Models Before Data Augmentation

Model	F1 Score	PR Curve		
		Car	Dog	Person
YOLOv5	81%	83	63	82
YOLOv8	72%	88	57	85
YOLOv9	72%	88	57	85
YOLOv10	66%	86	42	81
YOLOv11	76%	91	53	88
YOLOv12	61%	80	27	79

Results After Data Augmentation

Table 2 presents the performance metrics for the same models after applying data augmentation techniques.

Table 2: Performance Metrics of YOLO Models After Data Augmentation

Model	F1 Score	PR Curve		
		Car	Dog	Person
YOLOv5	73%	86	58	77
YOLOv8	86%	92	75	87
YOLOv9	76%	85	72	77
YOLOv10	70%	83	50	76
YOLOv11	79%	91	64	85
YOLOv12	60%	80	44	74

Model Selection and Recommendation

Based on a comprehensive evaluation of the provided results, **YOLOv8** with data augmentation emerges as the best-performing model for the object detection task across the classes "car," "dog," and "person." It achieves the highest F1 score of 86% and an average

PR curve score of approximately 84.7 (calculated as the mean of PR values: 92 for car, 75 for dog, and 87 for person). This outperforms all other models in both F1 score and average PR curve metrics post-augmentation. Notably, YOLOv8 shows significant improvement with data augmentation, with the F1 score increasing from 72% to 86% and the PR score for the "dog" class improving from 57 to 75, indicating strong generalization capabilities.

Compared to other models, YOLOv5 achieves the highest F1 score before augmentation (81%), but its performance declines post-augmentation (73%). YOLOv11 also performs well post-augmentation (F1: 79%, PR average: 80.0), but it is surpassed by YOLOv8 in all metrics. YOLOv12 consistently underperforms, with the lowest F1 score (60% post-augmentation) and poor PR scores, particularly for the "dog" class (44).

For the purposes of a thesis and general object detection tasks, **YOLOv8** with data augmentation is recommended due to its superior performance, robustness to augmentation, and balanced detection capabilities across all classes. Its high PR scores for "car" (92) and "person" (87), along with the best "dog" score among all models (75), make it particularly suitable for applications requiring reliable detection across diverse classes. Additionally, YOLOv8's widespread use and support in frameworks like Ultralytics facilitate implementation and further experimentation, which are valuable for thesis work.

Discussion

The results indicate varying impacts of data augmentation on the performance of different YOLO models. Notably, YOLOv8 shows a significant improvement in F1 score from 72% to 86% after augmentation, with PR curve values improving across all classes, particularly for the "dog" class (from 57 to 75). In contrast, YOLOv5 and YOLOv12 exhibit a slight decline in F1 scores post-augmentation, suggesting that the augmentation techniques may not have been equally effective for all models. YOLOv11 maintains relatively stable performance, with a modest F1 score increase from 76% to 79% and improved PR curve values for the "dog" class. The "car" class consistently achieves high PR curve values across all models, indicating robust detection capabilities, while the "dog" class generally shows the lowest PR scores, suggesting challenges in detecting this class. These findings can guide the selection of appropriate models and augmentation strategies for specific object detection tasks in future work.

Recommendations for Thesis Development

To develop a comprehensive thesis based on these results, consider the following structure:

- **Introduction:** Discuss the importance of object detection in computer vision and introduce the YOLO family of models.
- **Literature Review:** Review prior work on YOLO models and the role of data augmentation in improving model performance.
- **Methodology:** Detail the experimental setup, including dataset description, augmentation techniques, and evaluation metrics (F1 score, PR curve).
- **Results:** Present the results as shown in Tables 1 and 2, with a focus on YOLOv8's superior performance as discussed in the Model Selection section.

- **Discussion:** Analyze the impact of augmentation, model-specific performance, and class-specific challenges, emphasizing YOLOv8s robustness and limitations of models like YOLOv12.
- **Conclusion:** Summarize findings, highlight the recommendation of YOLOv8, and suggest future research directions, such as exploring advanced augmentation techniques or newer YOLO variants.

Ensure to include statistical analysis (e.g., significance testing) to validate differences in performance and discuss the practical implications of selecting YOLOv8 for real-world applications.